

Respiratory Agent Architecture: Complications & Cross-Organ Correlations

Project: BioSwarm Twin (Multi-Organ Digital Twin Sandbox)

Role: Respiratory System Agent Developer

Overview

In the multi-agent architecture, the Respiratory Agent does not manually calculate oxygen levels or lung mechanics. The central numerical simulation (Pulse Physiology Engine) handles the physics. The Respiratory Agent acts as an expert observer that reads the physiological snapshot, detects complications, applies clinical evidence registries, and cross-communicates with the Cardiovascular, Renal, and Hepatic agents.

Part 1: Core Respiratory Complications (ENGINE_SIMULATED)

These are the primary failure states the physics engine natively simulates. The Respiratory Agent must constantly monitor the physiological snapshot and flag these conditions when thresholds are breached.

1. Hypoxia & Hypoxemia (Low Oxygen)

- **Snapshot Variables to Monitor:** Arterial Oxygen Saturation (SpO_2 dropping below 92%), Arterial Partial Pressure of Oxygen (PaO_2 dropping below 80 mmHg).
- **Hackathon Triggers:** Severe asthma attack (high airway resistance), baseline COPD, or excessive IV fluid administration causing pulmonary edema.

2. Hypercapnia & Respiratory Acidosis (High CO_2 & Acidic Blood)

- **Snapshot Variables to Monitor:** Arterial Partial Pressure of Carbon Dioxide ($PaCO_2$ rising above 45 mmHg), Blood pH (dropping below 7.35).
- **Hackathon Triggers:** Hypoventilation events, most notably during opioid (e.g., Fentanyl/Morphine) administration which suppresses the central respiratory drive.

3. Pulmonary Edema (Fluid in the Lungs)

- **Snapshot Variables to Monitor:** Decreased lung compliance, rapid shallow breathing (tachypnea), dropping SpO_2 .
- **Hackathon Triggers:** Cardiovascular system failure or fluid overload (excessive IV saline).

4. Bronchoconstriction / Airway Obstruction

- **Snapshot Variables to Monitor:** Increased Airway Resistance, decreased Tidal Volume.
- **Hackathon Triggers:** Baseline COPD or injected stressors like Anaphylaxis or Asthma.

Part 2: Cross-Organ System Correlations (Agent Debate Matrix)

Organs do not exist in isolation. The Respiratory Agent must send and receive state messages to the Cardio, Renal, and Hepatic agents to form systemic conclusions.

Respiratory ↔ Cardiovascular (The Cardiopulmonary Axis)

- **When Cardio Fails, Respiratory Drowns:** If the Cardiovascular agent detects Heart Failure and the user administers 2 Liters of IV Saline, the heart cannot pump it. Fluid backs up into the lungs.
 - **Action:** Flag an ENGINE_SIMULATED finding of *Pulmonary Congestion/Edema* and explicitly state that the SpO_2 drop is secondary to heart failure, not a primary lung defect.
- **When Respiratory Fails, Cardio Strains:** If the patient has COPD, pulmonary blood vessels constrict (Hypoxic Pulmonary Vasoconstriction), forcing the heart to work harder.
 - **Action:** If chronic hypoxia is detected, send a warning message to the Cardiovascular Agent to monitor for Right-Sided Heart Strain (Cor Pulmonale).

Respiratory ↔ Renal (The Acid-Base Balancers)

- **The pH Tag-Team:** Blood pH is maintained by the lungs (fast, via CO_2 exhalation) and the kidneys (slow, via Bicarbonate HCO_3^- retention).
- **Action:** If the patient goes into Respiratory Acidosis (e.g., from an opioid overdose), the Respiratory Agent flags the high CO_2 . It must then check the Renal Agent's state. If the patient has Chronic Kidney Disease (CKD), the Renal Agent will flag an inability to compensate. Combine these for an EVIDENCE_ONLY warning: *Uncompensated Acidosis due to Dual Organ Impairment*.

Respiratory ↔ Hepatic (The Clearance Bottleneck)

- **The Opioid/Sedative Trap:** The Respiratory system is the victim; the Hepatic system is the culprit.

- **Action:** When opioids are given, respiration drops. If the Hepatic Agent flags "Severe Cirrhosis," the Respiratory Agent must generate an EVIDENCE_ONLY finding:
Extended Respiratory Depression Risk — hepatic clearance of opioid is impaired; hypoventilation will persist beyond standard biological half-life.

Part 3: Handling Systemic Diseases (Context & Guardrails)

When the simulation includes systemic conditions, the Respiratory Agent must apply contextual rules to the raw numerical physics.

1. Type 2 Diabetes Mellitus

- **Context:** Diabetic patients are prone to Diabetic Ketoacidosis (DKA), making the blood highly acidic.
- **Agent Logic:** Monitor dropping blood pH and check the respiration rate. The engine will simulate "Kussmaul breathing" (deep, rapid breathing to blow off CO_2).
- **Output:** EVIDENCE_ONLY - *Compensatory Hyperventilation (Kussmaul pattern) detected in response to suspected metabolic acidosis.*

2. Severe Sepsis / Septic Shock

- **Context:** Sepsis causes systemic inflammation, attacking lung capillaries and leading to Acute Respiratory Distress Syndrome (ARDS).
- **Agent Logic:** If systemic blood pressure plummets and sepsis is flagged, the Respiratory Agent must issue a proactive warning before fluids are given.
- **Output:** EVIDENCE_ONLY - *Sepsis-induced capillary leak suspected. High risk of ARDS. Avoid aggressive fluid resuscitation despite low blood pressure to prevent alveolar flooding.*

Developer Action Items (Hackathon Logic Flow)

To code this agent, implement the following loop at every simulation time-step:

1. **Fetch:** Pull Respiration_Rate, SpO2, PaCO2, Tidal_Volume, and Blood_pH from the Engine's JSON snapshot.
2. **Evaluate:** Run data against clinical thresholds (e.g., if SpO2 < 0.90 -> trigger Hypoxemia).
3. **Listen:** Check the LangGraph state channel for incoming messages from the Cardio, Renal, or Hepatic agents.

4. **Publish:** Push findings to the UI, strictly categorizing them as ENGINE_SIMULATED (raw physics) or EVIDENCE_ONLY (clinical registry rules).